Shrestha Pandey

AI Systems Architect · Full-Stack Developer · Founder, Concierge · Engineering, VIT Bhopal (Year 2)

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PROFILE

I like problems where being wrong is expensive: security, simulation, and software that has to keep working when nobody is watching. This year that has meant a desktop app where AI writes the code and every change carries its own proof, a client-side vault that splits secrets between people you trust, a crowd-safety twin for mass gatherings, and two NASA Space Apps entries. I'm also the founder of Concierge, where I build full-stack and AI software. Across 12 projects: 610 commits, 87,629 lines of source and 1,577 test cases.

EXPERIENCE & EDUCATION

Founder, Concierge Present
Architecting and building full-stack software, intelligent AI systems and high-performance web applications.

NASA Space Apps Challenge 2026
Two entries. One finds moving objects in SPHEREx all-sky images and grades them against JPL's catalogue; the other compares lunar south-pole landing sites using JPL Horizons data.

Team Terminal 3, IEEE SMCS Search and Rescue Competition 2026
A three-person team with a faculty advisor. We built a modular Python system that separates video information extraction, mission planning, multi-robot control and victim detection.

VIT Bhopal Year 2
Coursework and projects centred on systems architecture, algorithmic problem solving and software dynamics.

SELECTED PROJECTS

Kakeya Records OSI Hackathon 2026 · AI for mathematics
An AI agent finds new best-known thin-triangle Kakeya constructions in an hour on a laptop, each certified in exact arithmetic. shrestha-droid.github.io/kakeya-records

Branch HQ Desktop app · AI engineering
AI writes the code, your auditors get the proof: every change reaches your repo as a pull request carrying the scan, tests, review, model and approver. shrestha-droid.github.io/branchhq

Zero-Trust Vault v2 Web app · Cryptography
A client-side vault that splits secrets into Shamir shards, passphrases and passkeys. No server ever sees a key.
www.zerotrustvault.in

SPHEREx Moving Sky NASA Space Apps 2026 · Astronomy
Finds moving asteroids and Planet X candidates in SPHEREx all-sky images, graded against JPL's catalogue. spherex-moving-sky.vercel.app

CLPS Lunar Mission Browser NASA Space Apps 2026 · Visualisation
Compares lunar south-pole landing sites by when the Sun and Earth are above the horizon. shrestha-droid.github.io/clps-lunar-mission-browser

Drishti (दृष्टि) Live demo · Traffic AI
An AI traffic-signal concept for Indian mixed traffic, with a simulator and a fixed-time twin to benchmark against. shrestha-droid.github.io/drishti-demo

Bheed Drishti (भीड़ दृष्टि) Digital twin · Crowd safety
A crowd-safety twin of a temple town: a fixed barricade plan against an AI controller, on the same crowd. shrestha-droid.github.io/Bheed-Drishti

RING0 Web app · Education
An interactive OS simulator for scheduling, paging, deadlocks and synchronization, with shareable scenarios. shrestha-droid.github.io/ring0

MEND Interactive explainer · WebAssembly
Scratch an image and watch a C++/WASM Reed-Solomon decoder repair it, byte by byte. mend-azure.vercel.app

NCII Shield Privacy engineering · Live demo
Client-side perceptual hashing and BK-tree matching for takedowns, with the victim confirming every match. shrestha-droid.github.io/ncii-shield

Friday Desktop assistant · Local-first AI
A local-first AI assistant that runs native system commands through a background Python daemon, behind a WebGL holographic interface.

Zero-Trust Vault (v1) Web app · Cryptography
The first vault: a client-side cryptographic app with Shamir key-shattering and an air-gap mode. Rebuilt as v2. vault-hazel-gamma.vercel.app

SKILLS

Languages: TypeScript, JavaScript, Python, C++

Web: React, Next.js, Vite, Three.js / WebGL

Backend: Node + Express, FastAPI, Supabase

Systems: Electron, WebAssembly, Simulation, NASA / JPL data

Security: WebCrypto, Argon2id, Shamir sharing, WebAuthn, CSP

Tooling: Vitest, GitHub Actions, Vercel